

MAJID ZAREMEHRJARDY, P.ENG

Alberta Environment and Protected Areas, Calgary, AB
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HIGHLIGHTS

- Extensive knowledge of Alberta regulations, policies and frameworks related to surface water and groundwater.
- Experienced in building groundwater models including contaminant fate and transport models and reactive transport modeling.
- Expert in applying statistical analysis and machine learning techniques in analyzing, reporting, and visualizing environmental pressures and risks in local and regional scales.
- Experienced with regulatory processes related to environmental cumulative effects.
- Professional Engineer (P.Eng) in good standing with APEGA since September 2024.

WORK EXPERIENCE

Surface Water Specialist

October 2021 - Present

Alberta Environment and Protected Areas (EPA), Planning Branch

- Leading the data collection and analysis of water quality across the Lower Athabasca and Oil Sands Monitoring (OSM) regions, bringing together 2.5 million measurements of water quality parameters.
- Leading the development of Water Quality Dashboard and Annual Reporting Dashboard for supporting interactive representation of water quality and regulatory requirements in Alberta.
- Supporting the development of annual water quality and management response reports for regulating surface water quality conditions in Alberta.
- EPA representative on multi-stakeholder initiatives related to water quality such as OSM Technical Advisory Committee (OSM TAC) and various Watershed Planning and Advisory Councils (WPACs).

Term Lecturer

2022 - Present

University of Alberta, Department of Earth and Atmospheric Sciences

- (2023-) Geoscience Data Analysis (EAS 405/505)
- (2022) Hydrologic modelling (EAS 456/555)

University Research Assistant

September 2020 - October 2021

University of Alberta

- Used Hydroscape and Bioscreen models to model chloride transport and biodegradation of BTEX components and designing remediation scenarios.
- Collaborated with Alberta Innovates Water Innovation Program (AI-WIP) and EPCOR on regional modelling of groundwater-surface water interactions in North Saskatchewan River Basin using SWAT-MODFLOW model.
- Supervised an M. Sc. student and a B. Sc. student on groundwater modelling and drafting research articles.

EDUCATION

M. Sc. in Earth and Atmospheric Sciences

University of Alberta (2018-2020)

M. Sc. in Water Resources Engineering

Amirkabir University of Technology (2015-2018)

B. Sc. in Civil and Environmental Engineering

Amirkabir University of Technology (2011-2015)

TECHNICAL SKILLS

- **Programming:** Python, R, MATLAB, Fortran, L^AT_EX.
- **GIS Applications:** ArcGIS, QGIS, Leaflet, HEC-GeoRAS, ArcSWAT.
- **Water Resources Management:** SUTRA, MODFLOW, HYDROSCAPE, BIOSCREEN, SWAT-CUP, SWAT-MODFLOW, HEC-HMS, HEC-Ras, MODSIM, EPANET, StormCAD, LINGO, SewerCAD, QUAL2E.
- **General Utilities:** Microsoft Word, Microsoft Excel, Microsoft Powerpoint.

MEMBERSHIPS AND CERTIFICATES

- The Association of Professional Engineers and Geoscientists of Alberta (APEGA) - current designation: Professional Engineer (P.Eng) *2021-Present*
- Certificate of completion: “Principles of Hydrology” course provided by Canadian Society for Hydrological Sciences (CSHS) *2019*

COMMUNITY AND VOLUNTEERING ACTIVITIES

- Tutoring a student diagnosed with Autism Spectrum Disorder *2020–2023*
- Certified Fire Warden and First Aider in the South Petroleum Plaza building *2023*
- Teaching Assistant (TA) on “Advanced Hydrological Modelling”, “Advanced Groundwater” and “Technical English” courses. *2017-2020*
- Certificate of Completion: Question, Persuade and Refer (QPR) training for suicide prevention, University of Alberta *2019*
- Member of charity fundraising group for underprivileged students (as a member of Students for Students non-profit organization), University of Alberta *2019*